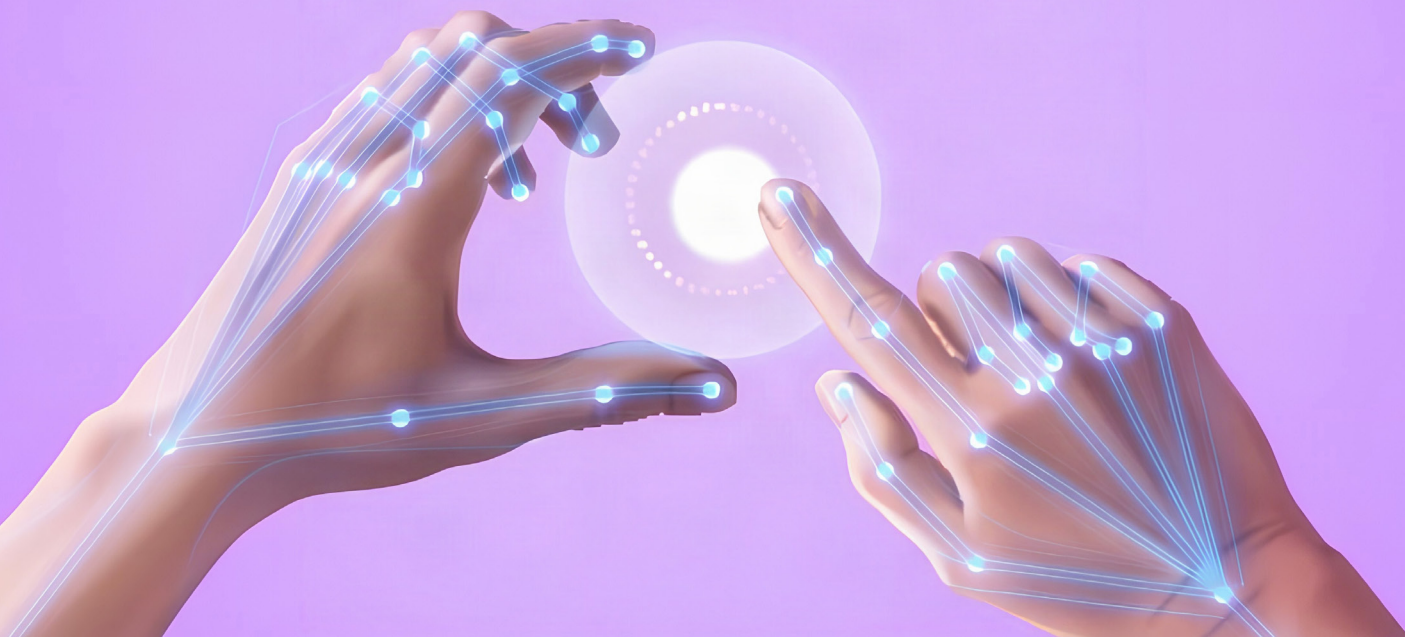


THE FUTURE OF SENSING: Smart Materials With Built-in Sensing



What if materials such as metal or plastic could feel what happens to them? Tiny built-in sensors now enable them to detect force and heat without changing how they function. The challenge now is manufacturing and scaling the technology for real-world use.



DR KONSTANTIN KLOPPE is the CTO at digid GmbH and provides tiny, advanced sensors that help materials become smarter and sense their surroundings.

Modern sensing systems typically rely on relatively large electronic sensors attached to objects or embedded as separate components. But what if, instead of relying on separate sensors to measure quantities such as force, pressure or temperature, everyday materials themselves could be engineered to become ‘smart,’ with built-in sensing capabilities that allow them to seamlessly integrate with the structures and environments they monitor?

This is achieved using ultra-miniaturised sensing elements that can be additively manufactured directly onto existing product components. These elements operate at a much smaller scale while consuming significantly less power than traditional sensing systems. The novel sensors can detect force and temperature while operating at nanowatt-level power

consumption, thanks to specially engineered high-resistivity materials.

Because the sensors are embedded into the material, its original mechanical properties remain unchanged. A metal part remains strong, while a plastic part remains flexible, but both can now also detect what is happening to them. For example, a metal beam can sense where it is being bent, while a surface can detect precisely where heat is increasing.

The technology is also designed to be practical for real-world use. It can be integrated using standard manufacturing methods and connected to existing electronic and IoT systems without requiring specialised infrastructure. It is also designed to minimise power consumption by transmitting only the most relevant information, reducing unnecessary data processing in remote and large-scale monitoring systems.